

Industrial Security (Bachelor)

Exercise Sheet – Section 10 Incident Response and Recovery

Exercise 10.1 – Write a Playbook

Draft a 1-page incident response playbook for:

“Suspicious unsolicited Modbus write commands detected from the corporate network.” Include:

- Roles and trigger conditions
- Detection inputs (IDS, firewall, historian)
- First-60-minutes response steps
- Containment options and their process impact
- Recovery and post-incident actions

Exercise 10.2 – Tabletop Exercise

In groups of six, play out the following scenario:

Saturday 03:00. The night-shift operator notices random valve cycling on the HMI. The Historian shows odd writes from an engineering workstation that should be powered off. The plant manager is unreachable.

Roles: operator, control engineer, OT security, IT security, plant manager, communications. Decide and document each action for the first three hours.

Exercise 10.3 – Forensic Walk-Through

Given a captured .pcap (provided by instructor) containing a Modbus injection attack:

1. Identify the attacker IP and the timestamp range of the attack.
2. Reconstruct the sequence of register writes.
3. Estimate the process impact based on the register tag descriptions provided.
4. Produce a 1-page forensic report suitable for sharing with management.

Exercise 10.4 – Business Continuity

For your water-treatment plant (Section 03):

1. Set RTO and RPO for each major system (SCADA, Historian, PLCs, HMI).
2. Identify which systems require cold spares.
3. Sketch a one-page manual-operations procedure for running the plant with no SCADA for up to 24 hours.

Exercise 10.5 – RACI for a Real Incident

Take the Modbus-injection playbook from Exercise 10.1 and turn it into a 6-row RACI for: operator, control engineer, OT security, IT security, plant manager, comms/legal. Each cell is one of R / A / C / I – justify your assignment of *Accountable* in two sentences.

Exercise 10.6 – Post-Mortem Draft

Pick any past incident (your own or a public one) and write a 1-page post-mortem using the 5-section template from the slides (timeline, what happened, what worked / what did not, action items, follow-up date). Keep the writeup blameless – focus on systems and decisions.